

Physics and Astronomy Department
The University of North Carolina at Greensboro
1400 Spring Garden Street
27412 Greensboro, North Carolina

February 16, 2026

Applying for the Position of Temporary Instructor of Physics

Dr. Pratap,

I write regarding the exciting opportunity to work a temporary instructor of Physics I and II at UNC-Greensboro, as advertised via announcement by Dr. Pratap. I am currently a doctoral candidate at UNC-Chapel Hill; I have completed all requirements for my degree but my dissertation, and anticipate defending my thesis in Spring 2027. I have defended a thesis to earn my Master's of Science in Physics, and while I remain a full-time graduate student at UNC, as a Royster Fellow with non-service funding I am available to fulfill the position this spring. My extensive experience as a physics instructor includes a long history of working with students enrolled in the aforementioned courses; I have acted as an assistant instructor at both Wake Forest and UNC-Chapel Hill in various roles across their respective introductory physics courses. My time teaching was formative, orienting me towards the efficacy of an interactive, experiential learning process where students develop and enhanced understanding of curricula via interactive modules. This experience in the classroom complements my eight years of experience as a one-on-one private physics tutor, which has provided me with insight into the learning processes of a diverse set of students, as I help them master the courses in question.

My educational experiences are buttressed by years of experimental biophysics research which gives me the applied knowledge base and technical expertise combined with broad philosophical study as an undergraduate which empowers me to answer questions with expertise and nuance. My works has given me a deeply grounded, functionalist approach to physics focused on how theory can be applied dynamically to better understand the world around us. As a result of these experiences and my undergraduate degree in philosophy, I have developed an academically broad, experimentalist perspective focused on how physics can be used as a tool to understand complex phenomena beyond the walls of the classroom. For instance, advised by Dr. Richard Superfine, I studied the stochastic processes of nanoparticles in poroelastic media during my Master's thesis, using the Ornstein–Uhlenbeck process, a technique typically associated with financial modelling, to better understand strain-driven changes in nanoparticle-matrix interactions. My history of applying physics in surprising ways to understand emergent phenomena prepares me to show students that no matter thier interests, physics can serve as a tool for advancing thier understanding.

Thank you for your time, and I look forward to talking with you about this exciting opportunity in the coming weeks.

Sincerely,

Alexander Marshall
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Alexander Marshall | Physicist

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🌐 philosophiaephysica.org

Royster Doctoral Fellow at UNC-Chapel Hill

Education

Doctorate of Philosophy in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2027

Royster Doctoral Fellow | Exams Passed | All But Dissertation

Master of Science in Physics

Chapel Hill, NC

The University of North Carolina at Chapel Hill

2025

Completed *en route* to PhD

B.S. in Physics, B.A. in Philosophy, Minor in Math.

Winston-Salem, NC

Wake Forest University

2022

GPA 3.91 | *Summa Cum Laude* | Stamps Scholar

Academic Projects

Honors Thesis.....

Generating a 3-D Lattice of Photonic Spheres to Track Fluorescently Labelled Chromatin

Advisor: Dr. Keith Bonin

Committee: Dr. Keith Bonin, Dr. George Holzwarth, Dr. Stephen Baker

Description: Wrote a custom iterative routine to generate, test, and analyze hundreds of thousands of phase settings on a spatial light modulator, implementing the correction method of Dholakia et al. (2017) Used this routine to create 3D lattice of photonic spheres to be used for tracking labelled chromatin in irradiated cells.

Masters Thesis.....

Physiological Strains in Poroelastic Media Induce Asymmetric Transport.

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. Greg M. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Kerry Bloom

Description: Tracked quantum dots undergoing dynamic diffusion in poroelastic media using LSM with unprecedented temporal resolution. Live imaging during physiological strain reveals that large solute transport is asymmetric during dynamic strains, with advection yielding significantly more forward angular anisotropy during the relaxation stroke than the compression stroke. These results suggest single-particle effects could explain the macroscopic solute concentration enhancements reported by studies of poroelastic media under dynamic strain.

Doctoral Thesis.....

Anomalous Diffusion in Poroelastic Media

Advisor: Richard Superfine

Thesis Committee: Dr. Richard Superfine, Dr. C.A. Forrest, Dr. David Hill, Dr. Amy Oldenburg, Dr. Ehssan Nazockdast

Description: Explore the role of static and dynamic strains in the transport of large solutes in various poroelastic media, including transcription factor search patterns in cellular nuclei exposed to dynamic strains.

Additional Research.....

Chalcogenide Crystals for Quantum Computing <i>Wake Forest Nanotech Center, Dr. David Carroll</i>	Winston-Salem, NC 2021
Neurochemical Addiction Mechanisms In the Nocus Accumbens <i>Wake Forest School of Medicine, Dr. Evgeny Budygin</i>	Winston-Salem, NC 2019-2021
Cultural Impact of Oenology in the Peleponnese <i>Wake Forest Department of Anthropology, Dr. Karen Friederic</i>	Winston-Salem, NC 2019
Cancer and Postoperative Pain <i>Wake Forest Baptist Medical Center, Dr. Christopher Peters</i>	Winston-Salem, NC 2018

Scholarships and Grants

2022-2027: Royster Doctoral Fellowship. Five-year fellowship awarded for academic merit.

2018-2022: Stamps Scholarship. Four-year scholarship awarded for academic merit.

2019: Wake Forest Scholars Travel Grant. Funded summer anthropology project abroad.

Technical skills

Coding Languages: MATLAB, LabVIEW, Python, L^AT_EX, Git

Scripting: μ Manager, ImageJ

Data Analysis: Particle Tracking, Image Analysis, Stochastic Processes

Other: Transmission Microscopy, Fluorescent Microscopy, Light Sheet Fluorescent Microscopy, Total Internal Reflection Microscopy, Custom Optical Design, Laser Systems, Spatial Light Modulation, Remote Focusing, Machine Vision, Atomic Force Microscopy, Misc. Lab Skills, Hazardous Waste Disposal

Presentations and Symposia

Generating a Three-Dimensional Lattice of Photonic Spheres <i>Symposium, Wake Forest University</i>	Winston-Salem, NC May 2022
Diffusion in Strained Poroelastic Media <i>Thesis Proposal, UNC-Chapel Hill</i>	Chapel Hill, NC September 16, 2025
Strain Alters Anomalous Diffusion in Poroelastic Media. <i>Symposium, Elon University</i>	Elon, NC October 1, 2025

Physiological Strains in Poroelastic Media Induce Asymmetric Transport. Chapel Hill, NC
Master's Defense, UNC-Chapel Hill November 17, 2025

Conferences

Undergraduate Research Day <i>Symposium, Wake Forest University URECA Center</i>	Winston-Salem, NC September 2019
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, Duke University</i>	Durham, NC September 15, 2025
Strain Driven Transport In Poroelastic Media <i>Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences</i>	Raleigh, NC September 23, 2024
Probing Structural and Force Dynamics during Phagocytosis <i>Triangle Cytoskeleton Meeting, NC Museum of Natural Sciences</i>	Raleigh, NC September 25, 2023
Various topics <i>Annual Colloquium, UNC Applied Physical Sciences Department</i>	Chapel Hill, NC 2022-2025
Strain-Driven Transport In Poroelastic Media <i>8th International Soft Matter Conference, Soft Matter Association of the Americas</i>	Raleigh, NC August 2024
Strain-Driven Transport In Poroelastic Media <i>Triangle Soft Matter Workshop, Duke University Dept. of Physics</i>	Durham, NC May 12, 2025
Strain-Driven Transport In Poroelastic Media <i>13th Carolina Biophysics Symposium, UNC-Chapel Hill</i>	Chapel Hill, NC November 6, 2025
Strain-Driven Transport In Poroelastic Media <i>Royster Global Conference, Royster Society of Fellows</i>	Chapel Hill, NC October 21, 2025
Strain-Driven Transport In Poroelastic Media <i>Biophysical Society Meeting, Biophysical Society</i>	San Francisco, CA February 21, 2026

Teaching and Mentorship

2021-Present: Assistant Instructor in various courses
2018-Present: Private tutor in advanced STEM fields.
2023-Present: Undergraduate Research Mentorship

Outreach

Hospital Volunteer <i>Wake Forest Baptist Medical Center, over 300 accumulated volunteer hours.</i>	Winston-Salem, NC 2015–2017
Open Flexure Microscope and Molecular Stamper <i>UNC-Chapel Hill Dept. of Applied Physical Sciences</i> Constructing low cost microscopes for use in low-resource environments. Project has recieved multiple awards, including the UNC MakerFest Entrepreneurship Award. Initiatives planned for building microscopes in low-resource highschoools in Eastern NC.	Chapel Hill, NC 2023-Present
UNC-Chapel Hill <i>UNC Science Expo</i>	Chapel Hill, NC 2023-Present

Annual outreach day full of public-facing science demos targeted as a young audience.

References

Richard Superfine, PhD <i>superfine@unc.edu, (919) 432-4381</i> Taylor-Williams Distinguished Professor, UNC-Chapel Hill	Doctoral Advisor <i>2022-Present</i>
Kerry Bloom, PhD <i>kerry_bloom@unc.edu, (919) 962-1182</i> Thad L. Beyle Distinguished Professor, UNC-Chapel Hill	Reasearch Collaborator <i>2025-Present</i>
David Hill, PhD <i>dbhill@email.unc.edu, (919) 962-4718</i> Associate Professor, UNC-Chapel Hill	Doctoral Committee Member <i>2025-Present</i>
Keith Bonin, PhD <i>bonin@wfu.edu, (336) 758-4962</i> Vice Provost, Wake Forest University	Undergraduate Advisor <i>2019-2022</i>

Languages

English: Fluent	<i>Native Speaker</i>
Modern Greek: Intermediate	<i>Conversational speaker.</i>
Classical Latin: Intermediate	<i>Seven years of formal instruction</i>